

Part 1: 15 MCQ

Part A: Matching Subfields of AI (Questions 1–5)

Instructions: Match each AI subfield (A–E) to the most fitting example/description (1–5).
Subfields:

- A. Natural Language Processing (NLP)
- B. Computer Vision
- C. Robotics
- D. Expert Systems
- E. Reinforcement Learning

Examples/Descriptions:

1. A system that processes and understands human text or speech to perform tasks such as translation or sentiment analysis.
2. A rule-based system designed to emulate a human specialist’s decision-making in areas like medical diagnosis.
3. A system that learns by trial and error, receiving rewards or penalties for certain actions in an environment.
4. A system that enables machines to interpret and analyze visual information from images or videos.
5. An autonomous machine capable of navigating and interacting with the physical world through sensors and actuators.

Part B: Multiple-Choice Questions (Questions 6–15)

6. Which of the following is a common challenge when deploying AI systems in real-world environments?
 - A. Lack of available frameworks
 - B. Limited hardware resources for GPUs/TPUs
 - C. Handling noisy, unstructured, or insufficient training data
 - D. Industry-wide ban on AI-based solutions
7. Which of the following best describes the concept of “transfer learning” in AI?
 - A. Training an AI model solely on synthetic data with no real-world input
 - B. Using a pre-trained model on one task and adapting it to a new, related task
 - C. Employing rule-based systems to solve general AI problems
 - D. Using generative models for data augmentation
8. In AI ethics, the principle of “fairness” typically refers to:

- A. Ensuring there are no data preprocessing steps
 - B. Guaranteeing that no one can reproduce the model's results
 - C. Creating AI systems that do not discriminate against protected groups
 - D. Making all AI systems open-source
9. Which of the following is not typically considered a key component of an AI “intelligent agent”?
- A. Ability to learn or reason about data
 - B. Observing the environment through sensors
 - C. Acting on the environment through actuators
 - D. Guaranteed zero error on every decision
10. In a typical supervised learning pipeline for image classification, which step is commonly performed after loading and labeling your dataset?
- A. Deploying the final model to a cloud environment
 - B. Data augmentation (e.g., rotations, flips, normalization)
 - C. Setting the learning rate to zero
 - D. Combining your training and test sets into one big set
11. In deep learning frameworks (e.g., PyTorch, TensorFlow), the function `backward()` (or equivalent) does which of the following?
- A. Computes and updates each weight matrix to its optimal value in one step
 - B. Automatically calculates gradients of the loss function w.r.t. each trainable parameter
 - C. Preprocesses input data by normalizing or standardizing features
 - D. Randomly initializes weights and biases before training

12. Consider the function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

(the sigmoid activation). Let $z = w_1x_1 + w_2x_2 + b$. Which expression correctly represents the partial derivative of $\sigma(z)$ with respect to w_1 ?

- A. $\frac{\partial \sigma(z)}{\partial w_1} = \sigma(z) [1 - \sigma(z)] x_1$
 - B. $\frac{\partial \sigma(z)}{\partial w_1} = 1 - \sigma(z)$
 - C. $\frac{\partial \sigma(z)}{\partial w_1} = x_1$
 - D. $\frac{\partial \sigma(z)}{\partial w_1} = \sigma(z)^2$
13. Suppose you have a loss function

$$L(w_1, w_2) = \frac{1}{2} (y - y')^2,$$

where $y' = w_1x_1 + w_2x_2 + b$. To perform gradient descent, you need $\frac{\partial L}{\partial w_1}$. Which is correct?

- A. $\frac{\partial L}{\partial w_1} = (y - y')$
- B. $\frac{\partial L}{\partial w_1} = -(y - y') x_1$

- C. $\frac{\partial L}{\partial w_1} = -(y' - y) x_2$
 D. $\frac{\partial L}{\partial w_1} = 2(y' - y) w_1$
14. You have a binary classifier producing a predicted probability p of the positive class. The true label is $y \in \{0, 1\}$. One common loss function is the negative log-likelihood (cross-entropy). Which is the correct expression for the cross-entropy loss L for a single data point?
- A. $L = (y - p)^2$
 B. $L = y \ln(p) + (1 - y) \ln(1 - p)$
 C. $L = -[y \ln(p) + (1 - y) \ln(1 - p)]$
 D. $L = \ln(1 + e^{-p})$
15. Consider a Bayesian update scenario: We have a prior $P(\theta)$ about a parameter θ . After collecting data D , we form the posterior $P(\theta | D)$. Which statement best describes the Bayesian rule?
- A. $P(\theta | D) \propto P(D | \theta) * P(\theta)$
 B. $P(\theta | D) = P(\theta)$
 C. $P(\theta | D) \propto P(D | \theta) * P(\theta) * P(\theta | \theta)$
 D. $P(\theta | D) = \int_{\theta} P(D | \theta) d\theta$

Part 2: 4 Problems

Problem 1.

We are given four points on a plane:

$$a_1 = (1, 1), \quad a_2 = (2, 1), \quad a_3 = (1, 3), \quad a_4 = (6, 1).$$

In this problem, we will explore how to combine these points into $K = 2$ clusters using the K -means algorithm. The distance between two points $\mathbf{x} = (x_1, y_1)$ and $\mathbf{y} = (x_2, y_2)$ is given by the Euclidean distance:

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

(a)

Briefly describe the stages of the K-means algorithm.

In your description, indicate:

1. How the **initialization** of cluster centers takes place.
2. How the **assignment** of points to clusters is performed.
3. How each cluster center is **updated** based on the points assigned to it.
4. Under what condition the process **terminates**.

(b)

If you run the K-means algorithm several times with different initial positions of the centers, do you **always obtain the same clustering result**?

(c)

Perform **ONLY ONE** iteration of the K-means algorithm using the following initial cluster centers:

$$\mathbf{m}_1^{(0)} = (1, 1), \quad \mathbf{m}_2^{(0)} = (2, 1).$$

1. **Assignment step:**

- For each of the four points $\{a_1, a_2, a_3, a_4\}$, compute the distance to each initial center $\mathbf{m}_1^{(0)}$ and $\mathbf{m}_2^{(0)}$.
- Assign each point to the cluster whose center is closer.

2. **Update step:**

After the points have been assigned to clusters, compute the new coordinates of each center:

$$\mathbf{m}_j^{(1)} = \frac{1}{|\mathcal{C}_j|} \sum_{\mathbf{x} \in \mathcal{C}_j} \mathbf{x},$$

where $\mathcal{C}_j$ is the set of points belonging to the j -th cluster.

- Provide the updated coordinates $\mathbf{m}_1^{(1)}$ and $\mathbf{m}_2^{(1)}$.

Problem 2.

(a)

Let $\mathbf{x} = [x_1, x_2]$ be a 2-dimensional vector.

$\|\mathbf{x}\|_p$ is called the **p -norm of vector $\mathbf{x}$** .

The **p -norm** of a vector $\mathbf{x} = (x_1, x_2)$ in $\mathbb{R}^2$ is defined as:

$$\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p)^{\frac{1}{p}}.$$

All norms represent distance in some sense. For example, the **Euclidian Distance** is just a special case of **p -norm** with $p = 2$:

$$\|\mathbf{x}\|_2 = (|x_1|^2 + |x_2|^2)^{\frac{1}{2}} = \sqrt{x_1^2 + x_2^2}.$$

Consider three regions $\{\mathbf{x} \mid \|\mathbf{x}\|_p \leq 1\}$ (all points x_1, x_2 that satisfy $(|x_1|^p + |x_2|^p)^{\frac{1}{p}} \leq 1$) for $p = 1, 2, \infty$.

Task: draw each of these regions on the coordinate plane.

Hint:

- For the ∞ -norm ($p = \infty$):

$$\|\mathbf{x}\|_\infty = \max\{|x_1|, |x_2|\}.$$

Each “unit circle” $\|\mathbf{x}\|_p = 1$ creates its own shape on the plane.

(b)

Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be two 4-dimensional vectors:

$$\mathbf{x}_1 = \begin{bmatrix} -2 \\ 3 \\ 4 \\ -1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ -5 \end{bmatrix}.$$

Let S be the subspace in $\mathbb{R}^4$ spanned by these two vectors $\mathbf{x}_1$ and $\mathbf{x}_2$.

1. Find $\langle \mathbf{x}_1, \mathbf{x}_2 \rangle$. What can you say about the **dimension** of the subspace S ?
2. Find the angle θ between $\mathbf{x}_1$ and $\mathbf{x}_2$. Use the formula:

$$\theta = \cos^{-1}\left(\frac{\langle \mathbf{x}_1, \mathbf{x}_2 \rangle}{\|\mathbf{x}_1\| \|\mathbf{x}_2\|}\right).$$

Answers in the form $\cos^{-1}(a)$ are also accepted.

Are $\mathbf{x}_1$ and $\mathbf{x}_2$ **orthogonal**?

3. Construct an orthonormal basis for S .

Hint:

- To find the **projection** of a vector $\mathbf{x}$ onto a vector $\mathbf{y}$, use the formula:

$$\text{proj}_{\mathbf{y}}(\mathbf{x}) = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \mathbf{y}.$$

(c)

Consider the 3×3 matrix:

$$A = \begin{bmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

1. **Rank and determinant:**

- (a) Find $\text{rank}(A)$.
- (b) Find the determinant $\det(A)$.

2. **Eigenvalues and Eigenvectors:**

Eigenvalues are special scalars λ associated with a square matrix A such that there exists a nonzero vector $\mathbf{x}$ (called an *eigenvector*) satisfying

$$A\mathbf{x} = \lambda\mathbf{x}$$

This means that $\mathbf{x}$ is such special vector that keeps its direction and is only scaled after it is multiplied by matrix A .

The eigenvalues are found by solving the *characteristic equation*

$$\det(A - \lambda I_{3 \times 3}) = \det\left(\begin{bmatrix} (-2 - \lambda) & -4 & 2 \\ -2 & (1 - \lambda) & 2 \\ 4 & 2 & (5 - \lambda) \end{bmatrix}\right) = 0,$$

where $I_{3 \times 3}$ is the 3×3 identity matrix.

- (a) Write down the *characteristic equation* and find the eigenvalues of A by solving it (you will solve a cubic equation and get three eigenvalues $\lambda_1, \lambda_2, \lambda_3$).
- (b) Find one eigenvector of A by plugging the smallest eigenvalue into the equation $A\mathbf{x} = \lambda\mathbf{x}$ and forming a system of equations.

Problem 3.

Consider the following simple neural network with three inputs (x, y, z) . The network performs two operations:

- **Addition:** $q = x + y$.
- **Multiplication:** $f = q \times z$.

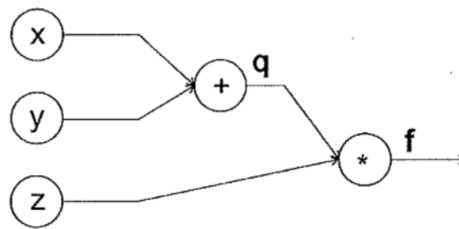


Рис. 1: **Diagram** (computation flow chart)

Let $x = -2$, $y = 5$, $z = -4$.

(a)

Compute q , then f : $q = x + y$, $f = q \times z$.

(b)

Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$.

(c)

Assume we have a function a **target value** f_{target} , which we want to get close to. We can define the loss function as:

$$L(x, y, z) = \frac{1}{2} (f(x, y, z) - f_{\text{target}})^2.$$

Our goal is to **minimize** $L(x, y, z)$, i.e., find such an x, y, z that $f(x, y, z) \approx f_{\text{target}}$.

Note: this can be done by calculating the partial derivatives of $L(x, y, z)$ and setting them to 0, but in this problem we want to familiarize you with **Gradient Descent**, which becomes useful when the function becomes more complicated than our $f(x, y, z)$.

Gradient Descent Step:

If η is the learning rate, a single step of gradient descent for x is given by:

$$x \leftarrow x - \eta \frac{\partial L(x, y, z)}{\partial x}.$$

To compute $\frac{\partial L}{\partial x}$, we can apply the **chain rule**:

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial f} \cdot \frac{\partial f}{\partial x}$$

Questions

1. Write down $\frac{\partial L}{\partial x}$ in terms of f_{target} , $f(x, y, z)$, and z .
2. Perform **one** gradient descent step and update x , y , and z with $\eta = 0.01$ and $f_{\text{target}} = -10$. When calculating, use the initial values of x , y , and z . What are the new values of x , y , and z ?
3. Calculate the new value of $f(x, y, z)$ using the new values of x , y , and z calculated in the previous subquestion.
4. Explain why following the gradient descent step formula is the same as bringing $f(x, y, z)$ closer to f_{target} ?

Hint:

- Partial derivative $\frac{\partial f}{\partial x}$ is used when we have a function of several variables but we assume that all of the variables are constants except for the one we are differentiating with respect to. For example, if $g(x, y) = x^2y + x$ is a function of two variables x and y , $\frac{\partial g}{\partial x} = 2xy + 1$ and $\frac{\partial g}{\partial y} = x^2$. When calculating $\frac{\partial g}{\partial x}$, we assume that y is a constant. When calculating $\frac{\partial g}{\partial y}$, we assume that x is a constant.

Problem 4.

You are provided with a dataset consisting of textual documents, categorized by sentiment. Each document is labeled as belonging to one of two sentiment categories:

- Negative Sentiment (denoted by “-”)
- Positive Sentiment (denoted by “+”)

The dataset is divided into two subsets: a training set and a test set. Your objective is to develop a model that can accurately predict the sentiment category of the documents in the test set based on the training data.

Task:

1. Using the Naive Bayes classifier, predict the sentiment category for each document in the test set.
2. State two weaknesses of the Naive Bayes classifier in a sentiment classification task.

Input Format: A table containing documents (text) and their sentiment labels (“-” or “+”).

Output Format: A list of predicted sentiment categories for each document in the test set.

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable with no fun

Рис. 2: Sample training/test dataset with corresponding sentiment labels.

Hint: To assist you in your calculation, refer to the following pseudo code for the Naive Bayes classifier, where V is the vocabulary (unique words) in the documents:

```
function TRAIN_NAIVE_BAYES(D, C) returns V, P(c), P(w|c)
    for each class c ∈ C # Calculate P(c) terms
        Ndoc = number of documents in D
        Nc = number of documents from D in class c
        prior[c] ← Nc / Ndoc

    V ← vocabulary of D
    bigdoc[c] ← append(d) for d ∈ D with class c

    for each word w in V # Calculate P(w|c) terms
        count(w, c) ← # of occurrences of w in bigdoc[c]
        total_word_count[c] ← sum of all word occurrences in bigdoc[c]
        likelihood[w, c] ← (count(w, c) + 1) / (total_word_count[c] + |V|)

    return prior, likelihood, V

function TEST_NAIVE_BAYES(testdoc, prior, likelihood, C, V) returns best c
    for each class c ∈ C
        score[c] ← prior[c]
        for each position i in testdoc
            word ← testdoc[i]
            if word ∈ V
                score[c] ← score[c] * likelihood[word, c]

    return argmaxc(score[c])
```

Рис. 3: Naive Bayes training and testing pseudocode.